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SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

PROJECT REPORT

Submitted for

HACK-O-WEEK PROGRAM

"**ContactHub:** *Smart Contact Book with Binary Search Visualization*"

A project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE AND ENGINEERING

Submitted By

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
AY 2025-26

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CERTIFICATE

This is to certify that the Project work entitled "**ContactHub**" is carried out by the **Swanandi Thakre, Girija Mahajan, Vansh Gaikwad,** in partial fulfillment for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, Symbiosis International (Deemed University), Pune during the academic year 2025-2026.

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DECLARATION

I hereby declare that the project titled "**ContactHub**" submitted to Symbiosis Institute of Technology, Constituent of Symbiosis International (Deemed University) Pune for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a result of original research carried out by me. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of any degree or diploma.

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Department: Computer Science and Engineering

Title of the project : **ContactHub**

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(Signatures of the All Students)

Date: 11.04.2026

ABSTRACT

ContactHub is an interactive, browser-based contact management application that demonstrates the practical implementation of fundamental data structures and algorithms. The project integrates an Array-based contact storage mechanism with the Binary Search algorithm to deliver an efficient and visually engaging experience for users managing their personal or professional contacts.

The application is built entirely using front-end web technologies — HTML5 for structure, CSS3 for styling, and JavaScript (ES6+) for logic and interactivity. Each contact entry stores a name, phone number, email address, and category (Friend, Family, Work, or Other). Contacts are automatically sorted alphabetically upon addition, ensuring the array remains in a sorted order required for Binary Search to function correctly.

The most distinctive feature of ContactHub is the step-by-step Binary Search Visualization panel, which animates the algorithm's decision-making process in real time — showing which element is being examined, which half of the array is being eliminated, and when the target contact is found. This makes the application not only a functional contact book but also an educational tool for understanding algorithm complexity and behaviour.

With a clean dark-themed UI, alphabet-based navigation, category tagging, and toast notifications, ContactHub delivers a polished user experience while effectively conveying the efficiency (O(log n)) of Binary Search over linear search methods. The project highlights how classical computer science concepts can be applied to solve real-world problems in an intuitive and accessible manner.

Keywords:

1. **Binary Search:** The core algorithm and primary technical focus.
2. **O(logn) Efficiency:** The mathematical proof of the app's performance.
3. **Visualization Panel:** The unique feature that animates the search process.
4. **Array-based Storage:** The fundamental data structure used.
5. **JavaScript (ES6+):** The primary language driving the logic.
6. **Alphabetical Sorting:** The prerequisite for the binary search functionality.
7. **Contact Management:** The real-world application of the code.
8. **Educational Tool:** The project’s secondary purpose for teaching complexity.
9. **Algorithm Complexity:** The theoretical concept being demonstrated.
10. **Interactive UI:** The front-end experience (dark-themed, toast notifications).

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CHAPTER 1: BACKGROUND AND TECHNICAL OVERVIEW

1.1 Introduction

In the modern era of computing, efficient data management and retrieval form the backbone of virtually every application. Whether it is a simple personal diary application or a large-scale enterprise customer relationship management (CRM) system, the ability to store, organize, and retrieve data quickly and reliably is paramount. Contact management applications represent one of the earliest and most universally understood categories of software, making them an ideal vehicle for demonstrating the application of data structures and search algorithms.

ContactHub is a web-based contact management system developed during Week 15 of the Hack-o-Week program. The project focuses on the practical implementation of two foundational computer science concepts: Arrays as a data structure and Binary Search as a search algorithm. By embedding these concepts into a real-world application, the project bridges the gap between theoretical algorithmic knowledge and practical software development.

1.2 Data Structures: Arrays

An array is one of the most fundamental data structures in computer science. It is a collection of elements stored at contiguous memory locations, indexed from zero. Arrays provide O(1) time complexity for random access (accessing an element by index), making them highly efficient for read-heavy operations. In ContactHub, each contact — consisting of a name, phone number, email address, and category — is stored as an object in a JavaScript array.

The contacts array is maintained in a sorted state at all times. Whenever a new contact is added, the array is re-sorted alphabetically using JavaScript's built-in Array.prototype.sort() method with a locale-aware comparator (locale Compare). This ensures that the array is always ready for Binary Search without requiring a separate pre-processing step.

1.3 Binary Search Algorithm

Binary Search is a highly efficient search algorithm that operates on sorted arrays. It works by repeatedly dividing the search interval in half. If the target value is less than the middle element, the search continues in the left half; if it is greater, the search continues in the right half. This process continues until the element is found or the search interval becomes empty. The time complexity of Binary Search is O(log n), making it significantly faster than linear search (O(n)) for large datasets.

In ContactHub, the binarySearch() method is implemented from scratch without relying on any library functions. It tracks each step of the algorithm — including which index is being examined, what comparison is made, and which half is eliminated — and stores these steps in an array. These steps are then rendered visually in the Binary Search Visualization panel with animated step cards.

1.4 Technology Stack

Technology	Purpose	Version
HTML5	Document structure and semantic markup	HTML 5.0
CSS3	Visual styling, animations, responsive design	CSS Level 3
JavaScript	Application logic, data structures, DOM manipulation	ES6+ (2015+)
Font Awesome	Icon library for UI elements	6.5.1
Google Fonts	Typography (Nunito & JetBrains Mono)	Web API

1.5 Architecture Overview

The application follows a clean single-page application (SPA) architecture with three separate files: week_15.html for structure, 15_stylecc.css for presentation, and 15_script.js for application logic. The ContactBook class encapsulates all data operations, including adding, removing, sorting, and searching contacts. The UI layer interacts with this class through a set of event-driven handler functions.

CHAPTER 2: PROBLEM STATEMENT AND MOTIVATION

2.1 Problem Statement

Traditional contact management applications provide basic CRUD (Create, Read, Update, Delete) functionality but rarely expose the underlying data organization or search mechanisms to the user. Students learning data structures and algorithms often struggle to connect theoretical concepts — such as sorting and searching — to real-world applications because educational tools are typically abstract and disconnected from practical software.

There is a need for an application that simultaneously serves as a functional contact manager and an interactive educational tool that makes the Binary Search algorithm transparent, step-by-step, and visually understandable. Existing contact applications treat search as a black box, offering no insight into how a name is located within potentially thousands of records. This opacity hinders learning and appreciation of algorithmic efficiency.

2.2 Motivation

The motivation behind ContactHub stems from the following observations:

- Students often find it difficult to visualize how Binary Search operates on real data, as textbooks typically present it in abstract pseudocode or on numbered arrays.
- Contact books are universally familiar, making them an ideal context for demonstrating search algorithms without requiring the user to learn a new domain.
- Most web-based contact managers lack any educational dimension — they do not show the algorithm behind the search, nor do they maintain visible sorted order.
- The gap between theoretical algorithm courses and practical web development skills needs to be bridged by projects that combine both worlds.

2.3 Objectives

The project sets out to achieve the following objectives:

- Implement a fully functional contact management system using an array-based data structure.
- Implement Binary Search from scratch and visualize each step of the algorithm in the browser.
- Maintain contacts in a sorted alphabetical order at all times to support Binary Search.

- Provide a clean, responsive, and dark-themed user interface that enhances the user experience.
- Enable users to add, delete, and search contacts with real-time visual feedback.
- Demonstrate the O(log n) efficiency of Binary Search compared to O(n) linear search.

CHAPTER 3: NOVELTY AND INNOVATIVE CONTRIBUTIONS

3.1 Step-by-Step Algorithm Visualization

The most novel aspect of ContactHub is its real-time, animated Binary Search visualization panel. When a user initiates a search, the application does not merely return a result — it animates each decision the algorithm makes. Each step appears as a styled card: purple cards indicate the element currently being examined, green cards signal a successful match, and dimmed red cards show eliminated halves. This level of transparency in algorithmic operation is rarely seen in practical web applications.

3.2 Live Sorted Array Maintenance

Unlike most contact managers that store entries in insertion order, ContactHub always maintains contacts in alphabetical sorted order. Every time a new contact is added, the array is re-sorted using JavaScript's locale-aware comparator. This is not only educationally significant (demonstrating the prerequisite of Binary Search) but also practically useful, as the contact list remains alphabetically organized at all times, improving browsability.

3.3 Deterministic Avatar Color System

ContactHub generates a consistent, visually distinct avatar for each contact based on the ASCII value of the first character of the contact's name. This deterministic approach — using charCodeAt(0) % colorPalette.length — ensures that the same contact always gets the same avatar color, creating a memorable and recognizable visual identity without requiring any image uploads or external services.

3.4 Alphabet-Based Scroll Navigation

The application features a 26-button alphabet navigation bar that allows users to jump directly to contacts starting with a specific letter. This secondary navigation mechanism complements the Binary Search feature and demonstrates a practical use of array traversal with findIndex(). If no contact starts with the selected letter, a toast notification informs the user, providing informative feedback.

3.5 Class-Based Object-Oriented Design in Vanilla JavaScript

The ContactBook class encapsulates all data and behavior in a clean object-oriented manner using ES6 class syntax — without any framework or library dependency. This design choice is pedagogically

significant as it demonstrates that clean, maintainable, object-oriented code can be written in plain JavaScript, making the project accessible and instructive to learners at any level.

CHAPTER 4: TECHNICAL ADVANTAGES AND PRACTICAL USEFULNESS

4.1 Algorithmic Efficiency

The Binary Search implementation in ContactHub offers O(log n) time complexity for search operations. For a contact list of 1,000 entries, Binary Search requires at most 10 comparisons to locate any contact, whereas a linear search would require up to 1,000 comparisons in the worst case. This represents a dramatic efficiency improvement that scales logarithmically with dataset size — a critical advantage in real-world applications with large user bases.

Contacts (n)	Linear Search (worst case)	Binary Search (worst case)
10	10 comparisons	4 comparisons
100	100 comparisons	7 comparisons
1,000	1,000 comparisons	10 comparisons
1,000,000	1,000,000 comparisons	20 comparisons

4.2 Zero External Dependencies

The entire application logic — including the data structure, sorting, searching, DOM manipulation, and UI rendering — is implemented in plain JavaScript without any framework such as React, Vue, or Angular, and without any external utility library such as Lodash or Underscore. This makes the project lightweight, fast-loading, and easy to deploy on any static hosting platform without a build step.

4.3 Responsive Design

The CSS employs a mobile-first responsive design using CSS Grid with auto-fill columns and media queries for screens below 768px. On smaller screens, the form fields stack vertically, the search button spans the full width, and the contact cards collapse to a single-column layout. This ensures a consistent, usable experience across desktops, tablets, and smartphones.

4.4 Educational Value

ContactHub serves dual purposes: it is both a functional application and an interactive learning tool. By making the Binary Search algorithm visible and animated, it helps students and developers intuitively understand why the algorithm is efficient, what decisions it makes, and how sorted data is a prerequisite for its operation. This visual pedagogy surpasses static code examples or textbook pseudocode in its educational impact.

4.5 Instant Feedback Mechanism

The toast notification system provides immediate, non-intrusive feedback for every user action — adding a contact, deleting a contact, sorting, and search outcomes (both found and not found). The notifications auto-dismiss after 2.5 seconds, ensuring the UI remains clean while keeping the user informed. This pattern follows modern UX best practices for single-page applications.

CHAPTER 5: DETAILED METHODOLOGY / SYSTEM ARCHITECTURE

5.1 System Design

ContactHub follows a client-side Model-View-Controller (MVC)-inspired architecture. The ContactBook class acts as the Model, encapsulating the data array and all operations on it. The HTML file defines the View — the DOM structure that the user sees. The JavaScript event handlers and rendering functions serve as the Controller, responding to user inputs and updating the View based on Model state.

5.2 ContactBook Class

The core of the application is the ContactBook class, which manages an internal JavaScript array called contacts. The class exposes the following methods:

Method	Description	Complexity
add(contact)	Pushes contact to array, then sorts	O(n log n)
remove(name)	Finds by name and splices from array	O(n)
sortContacts()	Sorts array using localeCompare	O(n log n)
binarySearch(name)	Binary search, returns steps array	O(log n)
getAll()	Returns a shallow copy of the array	O(n)
getByIndex(i)	Returns element at index i	O(1)
size()	Returns array length	O(1)

5.3 Binary Search Implementation

The binarySearch(name) method implements the classic iterative Binary Search algorithm. It initializes two pointers — left (0) and right (contacts.length - 1) — and iterates while left is less than or equal to right. At each step, it calculates the midpoint index and compares the contact name at that index (case-insensitively) with the search query. Based on the comparison, it either records a match, eliminates the left half, or eliminates the right half. Each decision is captured as a step object and returned as part of the result for visualization.

5.4 UI Rendering Flow

The render Contacts (highlight Index) function is responsible for all contact card rendering. It iterates over the contacts array and generates HTML for each card. If a highlightIndex is provided (from a successful Binary Search), the corresponding card receives the CSS class 'found', which applies a green glow border and background highlight. Cards animate into view using a CSS slideIn keyframe animation.

5.5 Search Visualization Flow

When the user triggers a Binary Search, the displaySearchSteps(steps, foundIndex) function receives the steps array from the ContactBook and uses setTimeout with incremental delays (800ms per step) to animate each step card appearing sequentially. This creates a live, step-by-step visualization that shows the algorithm's progress in real time, making the search process transparent and educational.

5.6 Data Flow Diagram

User Input (Form/Search) → Event Handler → ContactBook Method → Updated Array → renderContacts() → DOM Update → Visual Feedback (Toast / Visualization Panel). This unidirectional data flow ensures predictable state management without requiring any state management library.

CHAPTER 6: PRIOR ART AND RELATED WORK (LITERATURE SURVEY)

6.1 Binary Search – Foundational Research

Binary Search is one of the oldest and most studied algorithms in computer science, first formally described by John Mauchly in 1946. The algorithm's correctness was rigorously proven by D.H. Lehmer in 1960. Knuth's seminal work "The Art of Computer Programming, Vol. 3: Sorting and Searching" (1973) provides the definitive mathematical analysis of Binary Search, establishing its O(log n) time complexity and O(1) space complexity for iterative implementations.

6.2 Algorithm Visualization Tools

The field of algorithm visualization has a rich history. Systems such as BALSA (Brown Algorithm Simulation and Animation System, 1985), JSVEE (2013), and VisuAlgo (2011, NUS Singapore) have explored the pedagogical value of animating algorithms. These systems demonstrate that students who learn algorithms through visualization show significantly improved conceptual understanding compared to those learning through static text. ContactHub draws inspiration from this body of work by embedding visualization directly into a practical application rather than an isolated learning tool.

6.3 Contact Management Applications

Commercial contact management systems such as Google Contacts, Apple Contacts, and Microsoft Outlook all use highly optimized search mechanisms including B-trees and hash-based indexing. However, none of these systems expose their internal search logic to users. Open-source projects such as Nextcloud Contacts provide web-based contact management but similarly abstract away algorithmic details. ContactHub differentiates itself by making the search algorithm a visible and interactive feature.

6.4 Web-Based Data Structure Visualizers

Projects such as Data Structure Visualizations (University of San Francisco), Visualgo.net, and Algorithm Visualizer (GitHub) provide rich visualizations of data structures. However, these are purely educational tools without practical utility. ContactHub bridges the gap by combining practical functionality with algorithmic transparency, creating a tool that is useful for both everyday contact management and algorithmic education.

CHAPTER 7: APPLICATIONS AND DEPLOYMENT AREAS

7.1 Educational Institutions

ContactHub can be deployed as an interactive teaching aid in computer science classrooms at both secondary and undergraduate levels. During Data Structures and Algorithms courses, instructors can use the application to demonstrate Binary Search on real data, making the algorithm tangible and memorable. The visualization panel can serve as a live demonstration tool during lectures, replacing abstract pseudocode slides with interactive examples.

7.2 Personal Productivity

As a standalone single-page application with no server-side dependencies, ContactHub can be used directly in any modern web browser as a personal contact organizer. Users benefit from the alphabetically sorted display and the efficient search functionality. The category tagging (Friend, Family, Work, Other) adds organizational value for users managing contacts across multiple domains of their life.

7.3 Coding Bootcamps and Hack-o-Week Programs

The project is ideally suited for demonstration at hackathons, coding bootcamps, and skill-development programs such as Hack-o-Week. It showcases a complete application built entirely with front-end technologies, demonstrating proficiency in HTML, CSS, JavaScript, object-oriented programming, algorithmic thinking, and UI/UX design — all within a single, self-contained project.

7.4 Extensibility for Enterprise Applications

The ContactBook class can be extended to support server-side persistence (via REST APIs or local storage), advanced search filters (by category, phone number, or email), import/export functionality (vCard format), and synchronization with cloud contact services. The class-based architecture makes such extensions straightforward without requiring a rewrite of the core logic.

7.5 Algorithm Benchmarking Demonstrations

ContactHub can be extended with a performance comparison mode where both Linear Search and Binary Search are run on the same dataset simultaneously, with their step counts displayed side-by-side. This would make it an effective demonstration tool for benchmarking algorithmic efficiency and visually proving the superiority of O(log n) over O(n) at scale.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

ContactHub successfully achieves its goal of combining a functional web application with an interactive algorithmic learning tool. By implementing the Binary Search algorithm from scratch and visualizing each of its steps in real time, the project demonstrates how classical computer science concepts remain relevant and powerful in modern web development contexts.

The project reinforces several key learning outcomes: the importance of sorted data as a prerequisite for Binary Search, the dramatic efficiency advantage of O(log n) over O(n) searching, the practical implementation of object-oriented design in JavaScript, and the value of responsive and accessible UI design. All objectives set out at the beginning of the project have been met.

The technology choices — HTML5, CSS3, and vanilla JavaScript — ensure that the application is lightweight, portable, and accessible to any user with a modern web browser, without any installation or server infrastructure. This makes ContactHub not only a technically sound project but also a practically deployable application.

8.2 Future Scope

- LocalStorage / IndexedDB Integration: Persist contacts across browser sessions so that data is not lost upon page refresh, making the application genuinely usable for daily contact management.
- Search Comparison Mode: Display Linear Search and Binary Search running simultaneously on the same query, with a live step counter to visually demonstrate the efficiency gap.
- Import / Export (vCard): Support importing contacts from .vcf (vCard) files and exporting the contact list in standard formats compatible with Google Contacts, Apple Contacts, and Outlook.
- Edit Contact Feature: Allow in-place editing of existing contact details with inline form editing and optimistic UI updates.
- Multiple Search Algorithms: Extend the visualization to include other search strategies such as Linear Search, Jump Search, and Interpolation Search, allowing side-by-side algorithm comparison.
- Progressive Web App (PWA): Convert the application into a PWA with offline capability and home screen installation support for mobile devices.
- Backend Integration: Connect to a Node.js / Express or Firebase backend to enable multi-device synchronization, user authentication, and cloud-based contact storage.

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APPENDICES

Appendix A: Key Code Snippets

A.1 Binary Search Implementation

The following is the core Binary Search method from the ContactBook class:

Binary Search Logic (O (log n)):

1. **Initialize** low as 0 and high as array.length - 1.
2. **While** low is less than or equal to high:
    a. Calculate mid = floor((low + high) / 2).
    b. **If** array[mid].name equals targetName, **return** mid.
    c. **Else if** array[mid].name is less than targetName, set low = mid + 1.
    d. **Else**, set high = mid - 1.

Appendix B: Project File Structure

File	Description
week_15.html	Main HTML document — application structure and layout
15_stylecc.css	Complete stylesheet — dark theme, responsive grid, animations
15_script__1_.js	Application logic — ContactBook class, event handlers, rendering

Appendix C: User Interface (UI) Wireframes

Note: You would typically include screenshots or sketches here.

- **Main Dashboard:** Features the search bar at the top, a scrollable contact list in the center, and the "Add Contact" floating button.
- **Visualization Panel:** A dedicated section that appears during a search, showing a horizontal row of "cards" representing the array, with pointers for Low, Mid, and High markers.
- **Notification System:** Toast messages in the bottom-right corner confirming "Contact Saved" or "Search Completed."